

## **COURSE FOLDER**

### **MN573: SAFETY AND RISK MANAGEMENT IN MINES**

#### **1. Overview**

MN573: SAFETY AND RISK MANAGEMENT IN MINES. L-T-P:3-0-0 (3-Credit)

Teachers (Theory):

1. Prof. S. K. Palei

Teaching Assistants:

1. V. Srikanth
2. Vivek Singh

#### **2. Plan**

- Week 1-2: Source of risk and hazard in mines, safety performance measures.
- Week 3-4: Stochastic models for accident prediction
- Week 5: Cost of accident
- Week 6-7: Risk Management (risk analysis, risk assessment & risk control)
- Week 8-9: System engineering approach to risk and safety; Hazard identification techniques
- Week 10-12: Human behavioural approach in safety
- Week 13: Safety audit and control

#### **3. Weightage**

| Mid Semester | Term Paper and Assignments | Semester Exam. |
|--------------|----------------------------|----------------|
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#### **4. Books:**

##### **Text Books**

1. Heinrich H.W. "Industrial Accident Prevention" McGraw-Hill Company, New York, 1980.
2. Krishnan N.V. "Safety Management in Industry" Jaico Publishing House, Bombay, 1997.
3. Lees, F.P., "Loss Prevention in Process Industries", Butterworth publications, London, 2<sup>nd</sup> edition, 1990.
4. John Ridley, "Safety at Work", Butterworth and Co., London, 1983.

5. Kejriwal, B.K. Safety in Mines: A Survey of Accidents, Their Causes and Prevention, Lovely Prakashan, Dhanbad, 2002.

#### **Reference Books**

1. Guidelines for Hazard Evaluation Procedures, Centre for Chemical Process Safety, AIChE, 1992.
2. BerardinisSlote, L. J., Handbook of Occupational Safety and Health, John Wiley and Sons, New York, 1998.

#### **5. Term papers:**

1. Comparison of safety scenarios in Indian coal and non-coal mines
2. Investigation of accident caused by firedamp explosions – exemplified by an Indian case study
3. Safety Management Plan for Indian mines – Genesis, Preparation & Implementation
4. Historical Developments of Accident Causation Theories: Past, Present & Future
5. Hazard – Identification, Characterization & Analysis
6. Safety Culture – Concepts & Applications
7. Risk Assessment Strategies for Explosive Storage, Transportation & Use
8. Safety and Health Management System in India in the context of ISO 45001:2028 & IS 14489: 1018
9. Risks Involved in Opening of a Sealed-off Area: Strategies & Precautions
10. Cognitive Workload in Mining Operations: Concepts, Measurement and Safety Implications
11. Fatigue and its Impacts on Mine Workers' Safety
12. Emergency Preparedness Plan: Organisation and its function
13. Risk Assessment of haul road accidents involving HEMM in opencast mines
14. Safety Inspection and Safety Audit: Tools for Evaluation of Safety Performance
15. Fault Tree Analysis for Accident Prevention in Mines

#### **6. Tutorial: N. A.**

#### **7. Practical: N. A.**

#### **8. Examination papers (Attached)**